

Charotar University of Science and Technology
Education Campus, Changa – 388 421.

M Sc (Nano Science and Technology)
SEMESTER II EXAMINATION
NT 711: Elective-I: Biophysical Principle

Date: 01/05/2019

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Attempt all questions
2. Questions in Part A should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of Part B in separate answer sheets.

PART A

Total marks: 20

Q1. Choose the correct option and put ✓ mark in front of it:

1. Which of the following factors does NOT influence the process of diffusion?
(a) molecular size (b) concentration difference across the membrane
(c) thickness of the membrane (d) none of the above
2. Which of the following factors inhibits the transport of solutes across the biomembrane?
(a) Presence of ionophores (b) metabolic uncoupler
(c) hydrophobicity (d) all of the above
3. Secondary active transport and active transport of biomolecules follows same mechanism. True / False
4. Cellulose wall of the plant cell can be classified into which of the following?
(a) Impermeable membrane (b) Semipermeable membrane
(c) Permeable membrane (d) artificial membrane
5. Which of the following biomolecules CANNOT be transported through Na-dependent co-transport mechanism?
(a) glucose (b) aminoacids (c) hormones (d) none of the above
6. Internalization of insulin in a cell occurs via the process of -----?
(a) Reverse endosmosis (b) phagocytosis
(c) pinocytosis (d) receptor mediated endocytosis
7. Graham's law of diffusion relates rate of diffusion and density. True / False
8. Hydrostatic pressure in glomerular capillaries remains constant while it declines in other capillaries. True/False
9. The colloidal osmotic pressure is also known as :
(a) Hydrostatic pressure (b) Blood pressure
(c) Capillary pressure (d) oncotic pressure

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10. Diffusion rate is inversely proportional to temperature. True / False
11. Which of the following is the major difference between living semipermeable membrane and dead semipermeable membrane?
(a) Living membrane has a different lipid composition than the dead membrane
(b) living membrane can have selective diffusion while dead membrane cannot
(c) None of the above
(d) both of the above
12. The conditions under which the filtration and absorption are balanced in any organ is known as
a) Isothermal condition b) isovolumetric condition
c) Isomeric condition c) All of the above
13. Which of the following is anion exchange resin?
a) $-SO_3H$ b) $-COOH$ c) $-OH$ d) None of the above
14. The most frequently used method of determining liquid viscosity, is based on :
a) Time of retention b) Time of flow c) Melting point d) Specific gravity
15. Dynes/cm is unit of _____
a) Viscosity b) Adsorption c) Surface tension d) Hydrotropy
16. Colloidal particles possess the property of attracting and holding other substance on their surface. The substance that sticks on the surface of colloid is called _____
a) Adsorbent b) Cation c) Anion d) Adsorbate
17. Freundlich's adsorption isotherm may be written as:
a) $a=kC^{1/n}$ b) $a=kC^{1/v}$ c) $a=k^{1/n}C$ d) $a=k^{1/v}C$
18. Tangential force per unit area per unit velocity gradient is called as _____
a) Hydrostatic pressure b) Surface tension
c) Viscosity d) Coefficient of viscosity
19. "Substances which lower the surface tension become concentrated in the surface layer, whereas substances which increase surface tension, are more concentrated within the interior of the liquid than in the surface" This principle is given by:
a) Langmuir b) Freundlich c) Gibbs-Thomson d) Newton
20. The force of attraction between molecules of the same substance is called:
a) Adhesion b) Adsorption c) Absorption d) Cohesion

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Instructions to the candidates:

1. Attempt all questions
2. Write answers for Section I and Section II of Part B in separate answer sheets.

Part B

Total Marks: 50

Section I

- Q1 A Answer Any two:**
- a) Highlight the differences between diffusion and osmosis? 4
 - b) Explain any four factors affecting the phenomenon of diffusion 4
 - c) Explain the effect of concentration differences on diffusion. Define Permeability Constant and Permeability Coefficient and write the equations describing each. 4
- B Enlist different types of diffusion.** 2
- Q2 A Give the examples of active transport mechanisms operating in human bodies and briefly explain any one** 4
- B Explain using a neatly labeled diagram, how the phenomenon of osmosis, active transport and diffusion function in coordination in intestinal epithelium** 6
- OR**
- B Explain the following :** 6
- a) Reverse Osmosis
 - b) Vapour pressure
 - c) Osmotic pressure

Section II

- Q3 A Answer any two:**
- a) Explain the mechanical methods to determine osmotic pressure 4
 - b) Explain the physiological mechanism of osmoregulation in plants 4
 - c) Differentiate between Capillary filtration and Glomerular filtration 4
- B Enlist the factors affecting capillary filtration.** 2
- Q4 A Explain in detail sequence of events in phagocytosis using a diagram** 5
- OR**
- A Enlist the biological process that occur via the phenomenon of pinocytosis** 5
- B Briefly discuss Intestinal Absorption Vs. Absorption in the Alveoli** 3
- C Differentiate between (Any One)** 2
- a) Absorption and Adsorption
 - b) phagocytosis and pinocytosis
- Q5 A Answer Any Two**
- a) Write short note on factors influencing viscosity of Liquid 4
 - b) Explain method for Determination of Viscosity 4
 - c) Explain Biological Importance of Surface tension 4
- B Write short note on Capillary action** 2